

Rao Abdul Rehman

rao789rehman@gmail.com | +92-312-3882829 | github.com/Rehman-Maan | LinkedIn

Professional Summary

AI/LLM-focused Software Engineer with hands-on experience building production-style Django systems for RAG, document AI, support copilots, citations, evaluation dashboards, human review workflows, and deployment-ready backend infrastructure. Strong Python, Django, Celery, PostgreSQL, Redis, Docker, and OpenAI API foundation, supported by professional ERP/Odoo implementation experience and machine learning project work.

Technical Skills

- **AI/LLM Systems:** RAG, document question answering, cited answers, embeddings, vector search, LLM gateways, prompt workflows, local deterministic fallbacks
- **Backend:** Python, Django, Django REST Framework, Django Channels, Celery, Redis, PostgreSQL, pgvector, WebSockets, REST APIs
- **Document AI:** PDF/image processing, PyMuPDF, Pillow, OCR-oriented workflows, schema-based extraction, validation, human review, JSON/CSV exports
- **Evaluation & Safety:** retrieval recall, citation precision, faithfulness, groundedness, PII redaction, approval queues, audit trails, quality dashboards
- **DevOps & Testing:** Docker Compose, GitHub Actions, pytest, Playwright, Ruff, Trivy, Linux/SSH, environment-based settings
- **Additional:** Odoo, XML, QWeb, JavaScript, SQL, HTML, CSS, Pandas, NumPy, Matplotlib, YOLOv8, CLIP, CNNs

AI/LLM Portfolio Projects

Cited Knowledge Desk -- RAG Workspace with Grounded Citations 2026

- Built a Django-based RAG workspace for asking questions over uploaded documents with workspace-scoped document libraries, ingestion workers, semantic retrieval, streaming chat, and persistent source citations.
- Implemented PDF, Markdown, text, and DOCX support; Celery ingestion for text extraction and chunking; pgvector-compatible embeddings; and local/OpenAI provider switching.
- Designed citation records tied to source document, chunk, page, and match score, with focused excerpts so users can inspect the evidence behind every generated answer.
- Added answer feedback, reviewer queues, guest demo mode, and an evaluation harness tracking recall, citation precision, faithfulness, relevance, abstention, and latency.
- Packaged the project with Docker Compose, production deployment notes, GitHub Actions CI, Ruff, pytest, Playwright smoke tests, Docker build, and Trivy scanning.

Support Operations Copilot -- Safe AI Assistant for Ecommerce Support 2026

- Built an AI-assisted support operations workspace covering ticket triage, SLA risk, customer sentiment, escalation detection, thread summaries, and recommended next actions.
- Implemented retrieval-grounded reply drafting from support knowledge base content, with source snapshots, editable drafts, macro recommendations, and deterministic local fallbacks.
- Added safety controls including PII redaction before model calls, human-reviewed AI drafts, audit events, and lead approval queues for refunds, cancellations, credits, account closure, data export, and data deletion.
- Created a realistic ecommerce demo workspace with seeded tickets, customers, policies, approvals, screenshots, demo script, video assets, and evaluation dashboards.
- Built CI coverage for Django checks, migration checks, Ruff, unit/integration tests, Playwright smoke tests, static collection, support-eval smoke suite, Docker image build, and Trivy scan.

Multimodal Document Analyst -- PDF/Image Extraction and Review Workflow 2026

- Built a Django document AI workflow for invoices, receipts, forms, and scanned files, supporting PDF, PNG, JPG, and TIFF upload validation with workspace roles and document libraries.
- Implemented Celery-based processing using PyMuPDF for PDF text/page previews and Pillow for image normalization, then stored structured extraction runs with field confidence, source page, source text, and validation errors.
- Added configurable document types and extraction schemas, default receipt/invoice schemas, low-confidence human review queues, field correction, document approval, and audit events.
- Built cited document chat over processed page text, JSON/CSV exports with export history, and evaluation reports for classification accuracy, exact match, precision, recall, F1, missing fields, and invalid outputs.
- Included Docker Compose, production settings, deployment/security/testing docs, sample uploads, GitHub Actions CI, pytest, Playwright smoke tests, Docker build, and Trivy scanning.

Professional Experience

Odoo Technical Consultant, Odolution (Official Odoo Gold Partner), Karachi, Pakistan Sep 2024 -- Present

- Developed and customized Odoo 18/19 modules, reports, views, wizards, POS screens, receipts, website pages, portal flows, and role-based workflows for real client deployments.
- Migrated and aligned business logic across Odoo versions, including Odoo 14 to Odoo 18 custom modules, reports, views, security rules, and deployment checks.
- Built accounting, inventory, HR, approval, and reporting customizations using Python, PostgreSQL, XML, QWeb, JavaScript, Git, SSH, and Odoo.sh production workflows.
- Delivered dynamic Excel/PDF reporting, stock valuation/consumption logic, attendance automation, approval flows, landed cost enhancements, POS changes, and large export performance fixes.

Machine Learning Intern, Neurocomputation Lab (NCL), NEDUET, Karachi, Pakistan Mar 2023 -- Apr 2023

- Applied CNN models and transfer learning to classify Diabetic Retinopathy from fundus images, achieving 85%+ validation accuracy through preprocessing, augmentation, training, and evaluation.

Additional Machine Learning Project

Deep Learning-Driven Targeted Video Search for Personalized Object Identification 2024

- Built a personalized object detection system using YOLOv8, CLIP embeddings, and a Streamlit interface for text-based object search in live video, achieving 93% mAP@50 and approximately 30 FPS webcam performance.

Education

Bachelor of Engineering in Computer and Information Systems Engineering, NED University of Engineering and Technology, Karachi, Pakistan Nov 2021 -- Jun 2025

- **CGPA:** 3.46/4.00
- **Core Courses:** Machine Learning, Artificial Intelligence, DBMS, Software Engineering, OS, DSP, DSA, OOP

Achievements & Certifications

- Employee of the Month, Odolution: October 2025 and April 2026.
- Ranked 10th in Board of Intermediate Education Karachi with 95.27%; achieved 91.06% in Board of Secondary Education Karachi.
- Crash Course on Python, Coursera, August 2022 -- September 2022.
- Internship Certificate: ML/DL Techniques for Diabetic Retinopathy Classification, Neurocomputation Lab, NEDUET.

Languages

- Urdu: Native or Bilingual Proficiency
- English: Professional Working Proficiency